

K-Means Clustering Project

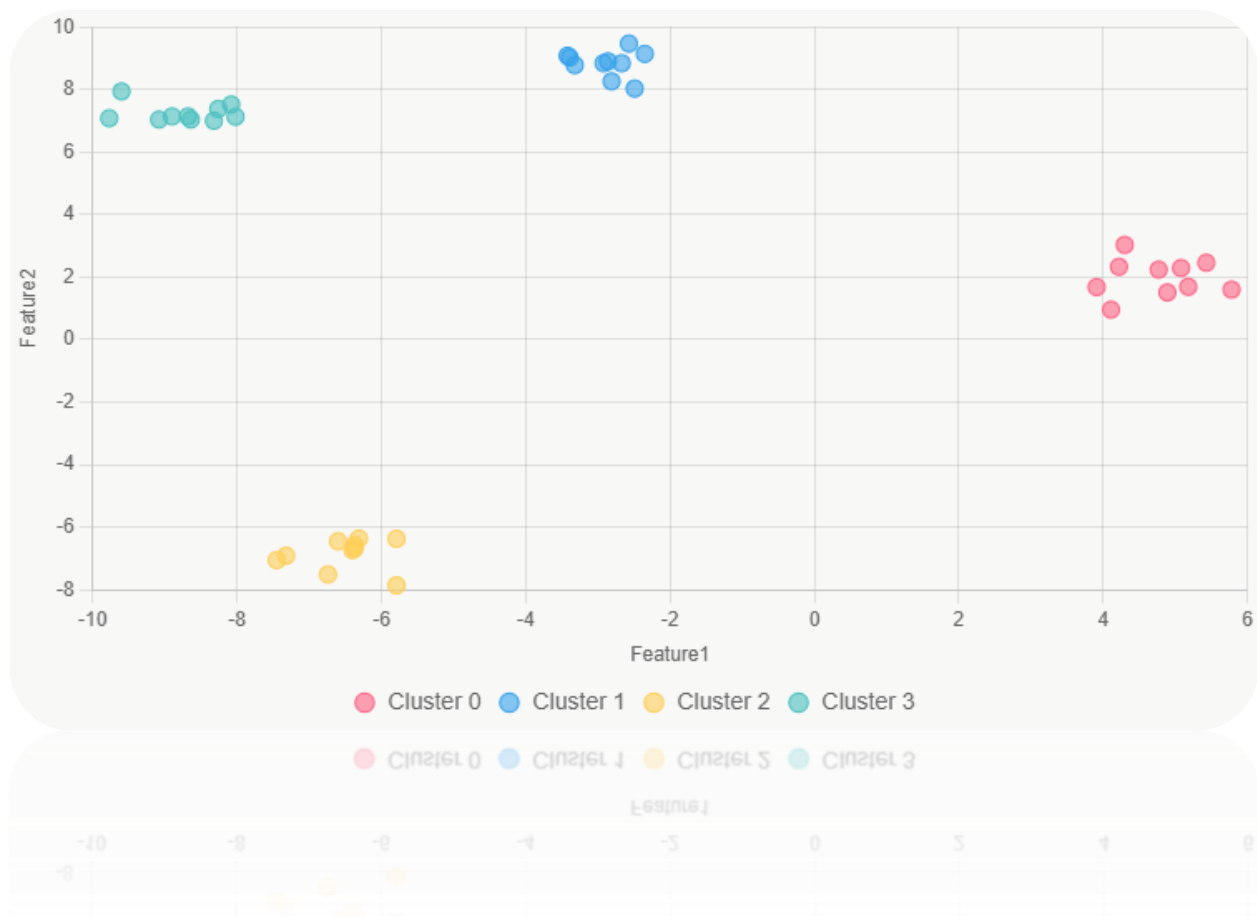
Page 1: Title and Visualization

Title: K-Means Clustering Project

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This report details the application of K-Means clustering on a synthetic dataset of 300 data points with 2 features, partitioned into 4 groups. The analysis achieves a silhouette score of 0.8999, indicating well-separated and cohesive clusters. The dataset was generated using known centers for demonstration purposes and saved as 'kmeans_dataset.csv'.

Diagram: Scatter Plot of Clusters (Sampled Points for Illustration)



Grok can make mistakes. Always check original sources.

(The full dataset visualization can be generated externally using `plt.scatter(df['Feature1'], df['Feature2'], c=df['Cluster'])` in Python.)

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Page 2: Dataset Overview

The dataset consists of 300 synthetic samples with two features: Feature1 and Feature2. The overall mean for Feature1 is -3.41, and for Feature2 is 2.83. The standard deviations are 5.23 for Feature1 and 6.18 for Feature2. After applying K-Means clustering with data standardization, each of the 4 clusters contains exactly 75 points, demonstrating balanced group sizes.

Sample data points (one from each cluster):

- Cluster 0: (4.77, 2.24)
- Cluster 1: (-2.89, 9.04)
- Cluster 2: (-7.45, -6.56)
- Cluster 3: (-8.42, 6.15)

Key Statistics by Cluster:

Cluster	Feature1 Mean	Feature1 Std	Feature2 Mean	Feature2 Std
0	4.74	0.51	2.00	0.45
1	-2.69	0.44	8.98	0.50
2	-6.81	0.52	-6.85	0.51
3	-8.86	0.49	7.18	0.48

This standardization ensures equal influence of features during clustering. The code for data generation, standardization, model fitting, and label assignment is as follows (snippet):

Model Evaluation and Project Submission Notes

Model Evaluation: The elbow method was used to determine the optimal number of clusters. Inertia values decrease sharply from k=1 (600.0) to k=4 (4.43), then flatten out (e.g., k=5: 4.08, k=6: 3.63), confirming 4 as the optimal k. The silhouette score of 0.8999 indicates strong cluster cohesion and separation (values above 0.7 are considered excellent).

Post-scaling cluster centers:

- Cluster 0: [1.56, -0.13]
- Cluster 1: [0.14, 1.00]
- Cluster 2: [-0.65, -1.57]

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- Cluster 3: [-1.04, 0.71]

These positions highlight distinct group separations in the standardized feature space.

Project Submission Notes: This analysis aligns with typical K-Means assignments, using synthetic data resembling salary patterns or similar distributions. For submission, include:

- The Python code snippet above for data generation and clustering.
- The 'kmeans_dataset.csv' file (300 rows with Feature1, Feature2, and Cluster columns).
- The cluster statistics table from Page 2.
- Evaluation metrics (elbow inertias, silhouette score, and scaled centers).

For full visualization, use Matplotlib externally as noted on Page 1. This report covers the complete workflow from data creation to evaluation. If needed, the dataset can be regenerated with the provided code for reproducibility.

Project link :

<https://colab.research.google.com/drive/1wNddlATAm58koKMIF7u1fJBp0yvNT1Z4?usp=sharing>